

Evaluation Metrics Report

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1. Introduction

The objective of this evaluation was to assess the performance of Natural Language Processing (NLP) models developed to rebuild the semantic understanding layer of the Social Engine. The labelled textual dataset was used to evaluate two classification tasks: sentiment classification and topic classification.

Multiple approaches were evaluated, including classical machine learning models, a lexicon-based sentiment method, and lightweight deep learning architectures. Model performance was assessed using accuracy, precision, recall, and F1-score, with particular emphasis on macro F1-score to account for differences in class distribution.

A consistent 75:25 train-test split was used for evaluation, with the test set kept separate from model development and tuning. Since the dataset contained repeated text entries, a group-aware split based on cleaned text was used to prevent identical texts from appearing in both training and test sets.

The evaluation ultimately identified TF-IDF with Logistic Regression as the selected trainable model for sentiment classification and TF-IDF with Linear SVM as the selected model for topic classification.

2. Dataset & Testing Setup

The evaluation was performed on Dataset 2, a labelled textual dataset provided for the NLP stage of the Social Engine task. The dataset contains 9,000 text records with four original attributes: text_id, post_text, sentiment_label, and topic_category.

Two classification tasks were evaluated:

- Sentiment Classification: Negative, Neutral, and Positive
- Topic Classification: Account_Security, Community_Discussion, Feature_Feedback, and Technical_Issues

The dataset was divided into 75% training data and 25% test data. This resulted in 6,773 training samples and 2,227 test samples.

Because the dataset contained repeated text entries, a group-aware train-test split based on cleaned text was used. This ensured that identical text entries did not occur in both the training and test sets. The resulting split contained 0 identical texts across the training and test sets.

The sentiment classes were evenly distributed, with 3,000 samples each for Negative, Neutral, and Positive. In contrast, the topic classes were highly imbalanced, with Community_Discussion accounting for 86.13% of the dataset. Therefore, evaluation of topic classification places particular importance on class-wise metrics and macro F1-score rather than accuracy alone.

The test set was kept separate from model tuning and was used only for the final performance evaluation reported in this document.

Dataset summary table

Parameter	Value
Total records	9,000
Original features	4
Training samples	6,773
Test samples	2,227
Train-test ratio	75:25
Sentiment classes	3
Topic classes	4
Identical texts across train/test	0
Sentiment distribution	3,000 per class
Dominant topic	Community_Discussion (86.13%)

3. Metrics Used and Why

The models were evaluated using Accuracy, Precision, Recall, Macro F1-score, and Weighted F1-score. These metrics provide complementary views of classification performance and are particularly important because the sentiment and topic tasks have different class distributions.

- Accuracy measures the proportion of test samples classified correctly. It provides an overall measure of correctness but can be misleading for highly imbalanced datasets.
- Precision measures how many of the samples predicted as a particular class actually belong to that class. It indicates the reliability of positive predictions for each category.
- Recall measures how many of the actual samples belonging to a class were correctly identified. It is useful for determining whether a model is missing samples from particular classes.
- Macro F1-score calculates the F1-score independently for each class and then gives equal weight to every class. This was treated as the primary comparison metric, particularly for topic classification, because the topic distribution is highly imbalanced. It prevents the performance on the dominant Community_Discussion class from masking weak performance on minority classes.
- Weighted F1-score calculates the F1-score for each class while weighting it according to the number of samples in that class. It therefore reflects overall performance while accounting for the class distribution.

For sentiment classification, accuracy and macro F1 were considered together because the three sentiment classes are equally represented. For topic classification, greater emphasis was placed on macro F1 and class-wise precision, recall, and F1-score because of the substantial class imbalance.

Metric interpretation: No single metric was used in isolation. The combination of overall accuracy, class-wise performance, and macro F1 provided a more complete assessment of model behaviour across both tasks.

Metrics summary

Metric	What it measures	Importance in this evaluation
Accuracy	Overall proportion of correct predictions	Overall correctness
Precision	Reliability of predictions for each class	Identifies false-positive issues
Recall	Ability to identify actual members of each class	Identifies missed classes
Macro F1	Equal-weighted F1 across classes	Primary metric for imbalanced topic classification
Weighted F1	F1 weighted by class support	Overall performance considering class distribution

4. Consolidated Model Comparison

This section should present the complete evaluation matrix first, followed by a short interpretation. Since this is an Evaluation Metrics Report, we should keep the discussion focused on measured performance, not detailed model-selection reasoning yet.

4.1 Overall Evaluation Matrix

Model	Task	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
TF-IDF + Logistic Regression (Baseline)	Sentiment	0.5712	0.5710	0.5707	0.5708	0.5713
Improved TF-IDF + Logistic Regression	Sentiment	0.5734	0.5731	0.5730	0.5730	0.5735
VADER	Sentiment	0.5819	0.5876	0.5809	0.5753	0.5762
Multinomial Naive Bayes	Sentiment	0.5712	0.5660	0.5649	0.5653	0.5654
Linear SVM	Sentiment	0.5447	0.5452	0.5436	0.5440	0.5442
Simple Deep Learning	Sentiment	0.5478	0.5589	0.5482	0.5470	0.5471
BiLSTM	Sentiment	0.5685	0.5746	0.5685	0.5693	0.5695
TF-IDF + Logistic Regression (Baseline)	Topic	0.8828	0.6575	0.3339	0.3723	0.8437
Improved TF-IDF + Logistic Regression	Topic	0.9273	0.8272	0.5796	0.6591	0.9187
Multinomial Naive Bayes	Topic	0.8572	0.2150	0.2500	0.2308	0.7970
Linear SVM	Topic	0.9349	0.8875	0.5760	0.6706	0.9210
Simple Deep Learning	Topic	0.3754	0.3118	0.4424	0.2475	0.4695
BiLSTM	Topic	0.8662	0.4875	0.4550	0.4577	0.8585

4.2 Key Observations

Sentiment Classification

- VADER recorded the highest values among the tested sentiment approaches for accuracy (0.5819), macro precision (0.5876), macro recall (0.5809), and macro F1 (0.5753).
- Improved TF-IDF + Logistic Regression produced a macro F1 of 0.5730, compared with 0.5708 for the baseline.
- BiLSTM achieved 0.5693 macro F1, while Simple Deep Learning achieved 0.5470.
- The Linear SVM achieved 0.5440 macro F1 for sentiment.

Topic Classification

- Linear SVM achieved the highest overall accuracy (0.9349) and macro F1 (0.6706) among the tested topic models.
- Improved TF-IDF + Logistic Regression achieved 0.9273 accuracy and 0.6591 macro F1.
- The baseline had substantially lower macro F1 (0.3723) despite an accuracy of 0.8828.
- Multinomial Naive Bayes and the deep learning approaches produced lower macro F1 values than the two stronger TF-IDF-based approaches.
- The difference between topic accuracy and macro F1 demonstrates the effect of the dataset's substantial topic-class imbalance.

4.3 Metric Interpretation

The results show that accuracy alone does not provide a complete picture of topic-classification performance. For example, the baseline topic model achieved 88.28% accuracy but only 0.3723 macro F1, whereas Linear SVM achieved 93.49% accuracy and 0.6706 macro F1. Therefore, the evaluation considers macro F1 together with accuracy and class-level metrics when interpreting model performance, particularly for the imbalanced topic classification task.

5. Sentiment Classification Results

Seven models were evaluated for sentiment classification: Negative, Neutral, and Positive.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
VADER	0.5819	0.5876	0.5809	0.5753	0.5762
Improved TF-IDF + Logistic Regression	0.5734	0.5731	0.5730	0.5730	0.5735
TF-IDF + Logistic Regression (Baseline)	0.5712	0.5710	0.5707	0.5708	0.5713
BiLSTM	0.5685	0.5746	0.5685	0.5693	0.5695
Multinomial Naive Bayes	0.5712	0.5660	0.5649	0.5653	0.5654
Simple Deep Learning	0.5478	0.5589	0.5482	0.5470	0.5471
Linear SVM	0.5447	0.5452	0.5436	0.5440	0.5442

5.1 Key Findings

- VADER achieved the highest sentiment accuracy (58.19%) and macro F1 (0.5753).
- Improved TF-IDF + Logistic Regression achieved 57.34% accuracy and 0.5730 macro F1.
- The improved Logistic Regression showed a small improvement over the baseline, from 0.5708 → 0.5730 macro F1.
- BiLSTM achieved 0.5693 macro F1, close to the Logistic Regression models.
- Multinomial Naive Bayes achieved 0.5653 macro F1.
- Simple Deep Learning and Linear SVM showed lower performance, with macro F1 of 0.5470 and 0.5440 respectively.
- Overall, sentiment classification remained a relatively challenging task, with the tested models achieving macro F1 scores around 0.54–0.58.

5.2 Sentiment Model Used

For the final NLP pipeline, TF-IDF + Logistic Regression was used as the trainable sentiment model. VADER was retained as a benchmark for comparison.

VADER was evaluated as a baseline lexicon-based method. Although it achieved slightly higher sentiment performance, the final system used TF-IDF + Logistic Regression because it is a trainable supervised model integrated with the overall NLP pipeline.

6. Topic Classification Results

Six models were evaluated for topic classification across four categories: Account Security, Community Discussion, Feature Feedback, and Technical Issues.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
Linear SVM	0.9349	0.8875	0.5760	0.6706	0.9210
Improved TF-IDF + Logistic Regression	0.9273	0.8272	0.5796	0.6591	0.9187
BiLSTM	0.8662	0.4875	0.4550	0.4577	0.8585
TF-IDF + Logistic Regression (Baseline)	0.8828	0.6575	0.3339	0.3723	0.8437
Simple Deep Learning	0.3754	0.3118	0.4424	0.2475	0.4695
Multinomial Naive Bayes	0.8572	0.2150	0.2500	0.2308	0.7970

6.1 Key Findings

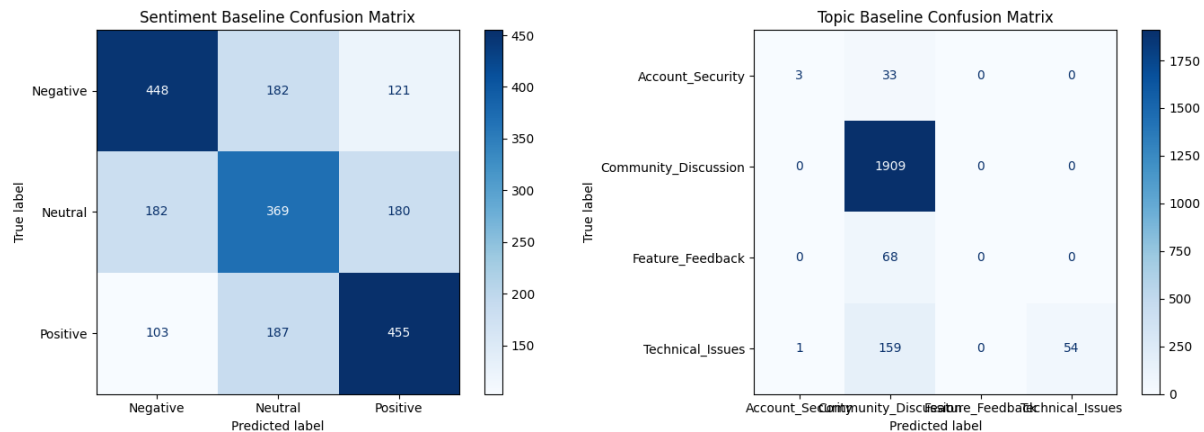
- Linear SVM achieved the highest accuracy (93.49%) and macro F1 (0.6706).
- Improved TF-IDF + Logistic Regression was close, with 92.73% accuracy and 0.6591 macro F1.
- The improved Logistic Regression showed a large improvement over the baseline: macro F1 increased from 0.3723 → 0.6591.
- BiLSTM achieved 0.4577 macro F1, lower than the two TF-IDF-based models.
- The Simple Deep Learning and Multinomial Naive Bayes models had much lower macro F1 scores.
- The difference between accuracy and macro F1 is important because the topic classes are highly imbalanced. Therefore, macro F1 gives a better view of performance across all topic classes.

6.2 Topic Model Used

TF-IDF + Linear SVM was selected as the final topic classification model because it achieved the highest accuracy and macro F1 among the tested topic models.

7. Confusion Matrices

Confusion matrices were used to understand how the baseline models classified each sentiment and topic class. Correct predictions appear on the diagonal, while off-diagonal values represent misclassifications.



7.1 Sentiment Confusion Matrix

The sentiment baseline confusion matrix shows:

- Negative: 448 correctly classified; 182 predicted as Neutral and 121 as Positive.
- Neutral: 369 correctly classified; 182 predicted as Negative and 180 as Positive.
- Positive: 455 correctly classified; 103 predicted as Negative and 187 as Neutral.

The main observation is that Neutral is frequently confused with both Negative and Positive. This indicates that distinguishing neutral posts from posts with clear sentiment is a major challenge.

7.2 Topic Confusion Matrix

The topic baseline confusion matrix shows:

- Community_Discussion: 1,909 correctly classified.
- Account_Security: only 3 correctly classified, while 33 were predicted as Community_Discussion.
- Feature_Feedback: none were correctly classified; all 68 were predicted as Community_Discussion.
- Technical_Issues: 54 correctly classified, while 159 were predicted as Community_Discussion.

The main observation is that the baseline model strongly predicts Community_Discussion, causing most minority-topic samples to be misclassified into this dominant class.

7.3 Key Findings

- Sentiment errors are mainly concentrated around the Neutral class.
- Topic errors are mainly caused by minority classes being classified as Community_Discussion.
- The confusion matrices support the earlier metric results and show why macro F1 is important, especially for topic classification.

8. Key Metric Observations

- Sentiment classification: VADER achieved the highest macro F1 (0.5753), followed closely by Improved TF-IDF + Logistic Regression (0.5730).
- Topic classification: Linear SVM achieved the highest macro F1 (0.6706) and accuracy (0.9349).
- Baseline improvement: Improved TF-IDF + Logistic Regression increased topic macro F1 from 0.3723 to 0.6591.
- Accuracy vs Macro F1: The topic results show a clear difference between accuracy and macro F1 because of the highly imbalanced topic classes.
- Classical ML vs Deep Learning: The TF-IDF-based classical models performed better than the tested Simple Deep Learning and BiLSTM models for topic classification.
- Sentiment difficulty: Sentiment models showed relatively close results, indicating that sentiment classification was more challenging to improve significantly.
- Overall: Macro F1 and class-wise metrics were important for evaluating model performance, particularly for the imbalanced topic classification task.

9. Final Selected Models and Metrics

The final models were selected based on their performance across the evaluated metrics and their suitability for the final NLP pipeline.

Task	Selected Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
Sentiment Classification	TF-IDF + Logistic Regression	0.5734	0.5731	0.5730	0.5730	0.5735
Topic Classification	TF-IDF + Linear SVM	0.9349	0.8875	0.5760	0.6706	0.9210

- Sentiment: TF-IDF + Logistic Regression was selected as the final trainable sentiment model.
- Topic: TF-IDF + Linear SVM was selected as the final topic model.
- Both models use the same TF-IDF-based text representation, making them suitable for integration into a single NLP pipeline.
- The final models were saved together in the trained model file for deployment and future prediction.